

Smart Home Automation Using ESP32

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Abstract

In this paper, a concept or idea for home automation is presented. It involves the use of an ESP32, Blynk, IR remote, and manual switch to control eight relays both online and offline while keeping an eye on the feedback in real time via the Blynk app. Device automation offers a lot of potential for both the current and upcoming generations. In the field of automation, mobile communication technology is playing a significant role. This home automation is based entirely on a low-cost, dependable home control monitoring solution that allows users of Android-based smart phones to access and manage devices and appliances remotely. Using this technology, the system creates a smart home by reducing human effort, increasing energy efficiency, and saving time. Additionally, it is very beneficial in helping to support disabled persons and meet their requirements at home so they may enjoy regular lives. This suggested system comprises of an Android phone running the ESP32 with the Blynk software, an IR remote control and manual control relays. Wi-Fi technology is being used to monitor the device because of its precision, long range and immediate connectivity. This module is very simple to install and easy to use, and it regulates the home appliances.

Keywords: ESP32, RELAY, LDR, IoT, BLYNK APP

INTRODUCTION

Considering that it is safer and more secure, home automation has grown more advantageous. Home automation has advanced and is now more accurate at monitoring all household appliances. Home automation technology have advanced into incredibly user-friendly and energy-efficient smart home innovations. It has fundamental features to preserve user comfort and happiness. Home automation is a special system that can manage and communicate with almost every component of your home. "Home automation" is the synchronization of all household amenities and equipment. For instance, a centrally located microcontroller panel may be able to operate all electrical appliances and heating systems. Home automation can involve centralized control of various appliances in our home through a single control centre, programming electrical devices to situations or scenarios, or operating

certain components of our home remotely using a computer or other mobile equipment. The many controllable appliances must be linked and in communication with one another. The primary function of home automation is to regulate or keep an eye on signals coming from various appliances or essential services. The home automation system may be controlled and viewed on a smartphone.

Smart home technology can play an important role to increase energy efficiency. Commonly, home appliances remain switched 'ON', due to the laziness or due to human negligence. Many time the home users leave the room without turning off the lights, fans, Air conditioners. Smart home automation systems can automatically 'switch

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OFF' home appliances without human intervention.

In this study, we suggested a low-cost, low-power Internet of Things-based system. The registered user can manage his or her household appliances with this system from any location at any time. Also, the registered person can monitor his/her home electrical appliances. We created a custom-made private server for monitoring and controlling the system. The server is communicated with the ESP32 Wi-Fi module.

The smart home with a cloud-based approach that provides secured real-time monitoring and controlling via the internet for home from a remote location also can be proven the best choice. This approach does not require any additional hardware or a huge memory to hold device data. The idea behind this home automation system is to integrate the mobile app with cloud networking using wireless communication. The users will use mobile applications as an interface with the cloud to monitor and control different appliances at home.

Blynk API is a Cloud-Platform that is used as iOS and android apps to control the microcontroller over the internet. It's a digital dashboard where a graphic interface for the project is built. It can control hardware remotely and it provides better secure communication. The Wi-Fi controller Esp32 provides a better range of communication and it is the successor of esp8266 with faster Wi-Fi and Bluetooth connection that runs at a typical frequency of 160 MHz. The IoT devices are generally responsive to augmented reality apps that are controlled through the internet and the status of these devices is updated through data exchange protocol. In this paper, the Augmented Reality app is built that controls the home appliances and it is also extended to multiple appliances like bulbs, fans, laptops etc. and the monitoring of the room's temperature is done.

LITERATURE REVIEW

Over times, a number of Smart houses with unique technology have emerged. Smart home automation is made possible by a rich blend of 3D items controlled by IoT and augmented reality. In the twenty-first century, the internet grew in popularity. A user-friendly web server designed to control the appliance's automation over MQTT. MQTT is an application layer protocol that displays sensor data, switch control information, and the concept of home automation utilizing IoT. The basic concepts of home automation are usually presented in this paper utilizing an intuitive web interface. For the purpose of using smart home automation, a product called IoT Home are developed [1]. They demonstrated the viability of Wi-Fi control using NodeMCU and showed that it can be utilized with Google Assistant and other monitoring tasks. Using wearable technology, it's possible to insert fictional objects that look real into the real world [2].

Bluetooth technology are employed for local interfacing and remote monitoring through the internet utilizing RTP (Real-time Transport Protocol) and a web-based GUI (Graphical User Interface) in a networked monitoring system for home automation. With a few small adjustments, the suggested solution was acceptable for several home automation applications. Using Bluetooth Low Energy, an approach to energy management for smart homes influence of high-power loads and standby appliances on consumer energy costs during peak hours are presented [3].

IoT applications can use augmented reality [4]. This study creates smart eyewear and augmented reality applications. The IoT application is rendered using smart glasses and a local server through communication between the AR application. This paper also provides evidence for IoT and AR. The Brain-Computer Interface, which uses eye blinks to operate household appliances via augmented reality, is based on brain signals sent through middleware and visual stimuli. It is possible to operate Internet of Things (IoT) devices via virtual reality using brain waves [5].

It is suggested to implement smart home automation utilizing opensource software Visual Basic, light and temperature sensors, and Arduino as the master controller [6]. With the use of a visual basic

software, voice instructions might be used to operate household equipment. Using the MATLAB GUI, the security system guarded the home against invaders. By including wireless communication and schedulers, the system could be improved.

Nowadays, there are more AR Mobile apps. the technique for controlling web-based mobile AR appliances in a 5G network. It promotes mobility, makes good use of the available internet, and offers a portable, cross-platform, and affordable end-to-end delivery solution. It also offers the consumers an immersive experience and notions about reality through virtual things [7].

Another work using Wi-Fi technology proposed an automation system providing power management and security in the home. The system was proposed for monitoring lights, temperature, and smoke and motion detection [8].

Users and system administrators have the option of controlling system code locally, via LAN (Local Area Network), or remotely, via the internet. However, a suitable web browser and the correct server I.P. Using Arduino and Wi-Fi, the Smart Home Automation and Security System included a web server with IP connectivity and an Android app for device remote control [9]. An LDR and a temperature sensor (LM35) has built inside the Arduino to provide automatic lighting and fan control. A dedicated server PC didn't need to be set up [10]. The system is capable of voice activation for switching operations, remote access to the home web server, user identification, device control and monitoring, scheduling, and automatic control of the home environment [11].

In this study, virtual buttons are used to interface with and control household appliances. It can be expanded to several targets and includes real-time monitoring of the room's temperature and humidity. The residents and smart devices should be educated to improve technology awareness and better quality of life required. The model employs the ESP32 module and includes a number of sensors, including a temperature sensor, PIR sensor, gas sensor, and water level sensor. The ESP32 transmits orders to cloud storage objects utilising an IoT. The cloud storage objects speak, receive commands, and then display a response [12].

SYSTEM DESIGN

The system's main controller, the ESP32, has a dual-core processor, built-in Wi-Fi, and compatibility for Bluetooth Low Energy.

The ESP32 microcontroller, the Arduino programming language, and an Android-based mobile application are used to construct the system shown in Figure 2. All electronic devices are integrated into one environment using the Microcontroller ESP32. The ESP32 also has 36 GPIOs, a Wi-Fi module, and a Bluetooth module. The ESP32 has a sizable memory. Additionally, ESP32 is suitable for use in projects to PIR sensors are employed to detect motion as illustrated in Figure 1.

HARDWARE DESCRIPTION

ESP 32

It is a low-cost, low-power platform on a chip microcontroller with built-in Bluetooth and Wi-Fi. The project's brain is the ESP32. All of the sensors are connected via a microcontroller board. To carry out the project's operations, the board is programmed using the source code. The ESP32's on-chip memory is where the source code is kept. The interaction between the coder and the user can be thought of as this block. It is therefore viewed as the project's core. The operating voltage range for the ESP32 is 2.2 to 3.6V. The ESP32 device will supply 3.3V of power to the chip during normal operation. Up to 240MHz of CPU frequencies and 512 kB of RAM are possible (Random-access memory). Also, it features numerous peripherals, including capacitive touch, ADCs, DACs, UART, SPI, and I2C. The board has two onboard buttons ENABLE and the BOOT button. It is an open-source resource and plug-and-play modules that support thousands of compatible boards and sensors. Every thing necessary to operate the microcontroller is present. The ESP32 pin description is shown in Figure 3.

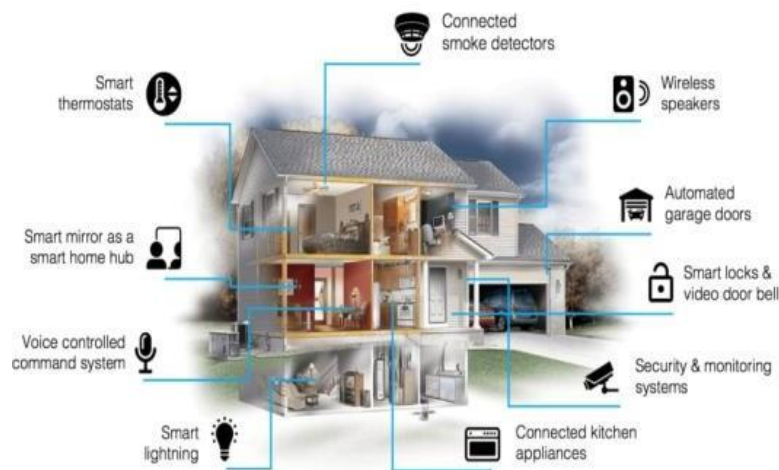


Figure 1. System Design of Home Automation.

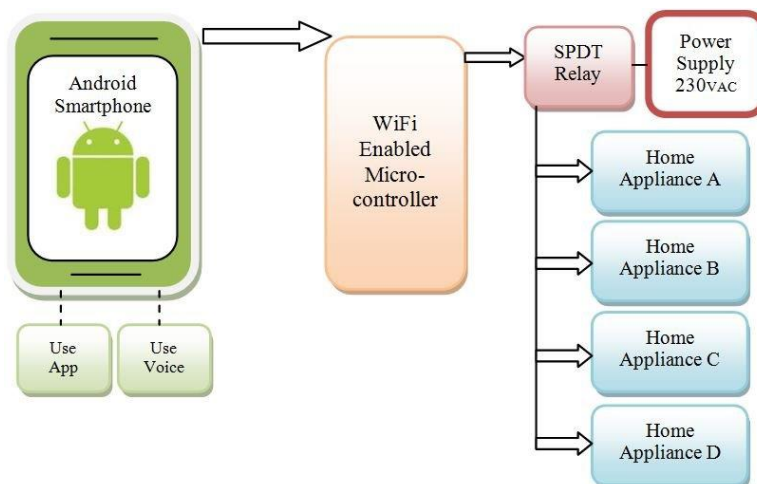


Figure 2. Block Diagram of Home Automation System.

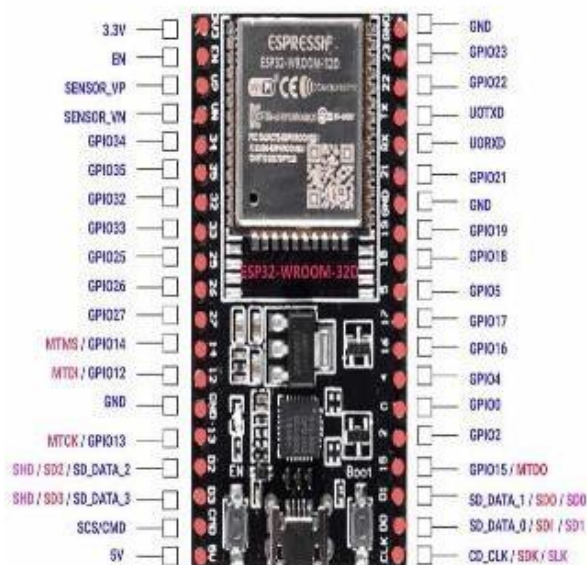


Figure 3. Pin Diagram of ESP-32 device.

Relay Module

Relay plays a significant role in this development and is used to manage high-voltage residential equipment. To control four appliances, we used a 5V four channel relay. This has four control inputs, VCC and GND, as well as four outputs. Four appliances are operated by the relay's outputs, which are coupled to four ESP 32 digital output pins through control inputs. Relay presented in Figure 4.



Figure 4. Relay Module.

Gas Sensor

A device that measures the existence or amount of gases in the atmosphere is called a gas sensor presented in Figure 5. By altering the resistance of the material inside the sensor, the sensor generates a corresponding potential difference based on the gas concentration, which may be recorded as output voltage. The type and concentration of the gas can be inferred from this voltage value.



Figure 5. Gas Sensor.

Rain Sensor Module

The rain device has a sensing pad with two nickel-coated copper series tracks as shown in Figure 6. Two header pins on this pad are fully developed to the pad's copper rails. These two header pins' primary function is to connect the Sensing Pad and the rain sensor module via two jumper wires. In this case, the sensing pad's one path receives a +5v power supply from the rain sensor module's one pin, while the pad's other pin supplies the return power.

Ultra Sonic Sensor

The transmitter and receiver are the two halves of an ultrasonic sensor as shown in Figure 7. These areas are close together to ensure that the sound travels directly from the transmitter to the intended target and back to the receiver. Calculation mistakes are minimized when there is little space between the transmitter and receiver sections.



Figure 6. Rain Sensor.



Figure 7. Ultra Sonic Sensor.

LDR Sensor

The photo conductivity principle drives the operation of this resistor. It is nothing more than the fact that when light strikes its surface, the material's conductivity decreases and the electrons in the device's valence band are stimulated into the conduction band. Each incident light photons' energies must exceed the band gap of the semiconductor material. Consequently, the electrons move swiftly from the valence band to the conduction band. Here one LDR sensor is depicted in Figure 8.



Figure 8. LDR Sensor.

METHODOLOGY

The demonstration of a virtual button-controlled home appliance using Blynk API and Unity3D is proposed and also to observe the real-time temperature sensor data. This paper mainly aims to give proof for the concept of Augmented Reality with IoT. The IoT part is controlling the home appliance using esp32 that is connected to relay and Blynk API through the internet.

In [13] The term "cloud computing" refers to the process of storing data in the cloud and using it to run connected implementations. The cloud, which is linked to several computers and servers over the internet, hosts everything.

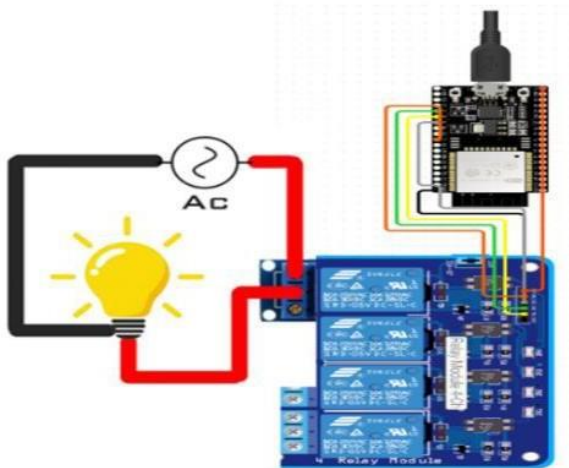


Figure 9. Connection Design of Home Automation System.

The IoT connection used to control an AC appliance is shown in the diagram above. One of the GPIO pins on the esp32 is used to connect it to the relay's input. Relay modules are switches that are controlled electronically and are used to turn on and off circuits. Relay modules are mostly used to handle voltages and currents that are greater than what a microcontroller is capable of handling. It uses a 5V relay module. It comprises of three terminal connections with COM, NC, and NO ports, together known as terminators [14]. VCC, GND, and IN pins are located on the other side and should be connected to the board as shown in Figure 9.

The relay module employs an electric current to open and close the switches with the assistance of a coil that pulls the switch contacts together when the coil is active and pushes them apart when it is not. The home appliance wire is divided in half, and the COM and NC ports are linked to these, with the VCC pin attached to the Vin pin of the ESP32. A new project is made in the Blynk Android app, and a button widget is added to it. Typically, the API key token used to control the appliance is sent to the mail by the Blynk app. The Arduino IDE is used to write the C++ code, which is then uploaded to the esp32. To control the Blynk device's virtual pin. The Blynk API key token is used in place of the auth token field, and V1 is used in place of the pin field (Virtual pin). And the value will be switched on and off at 0 or 1. The systematic work flow is presented in Figure 10.

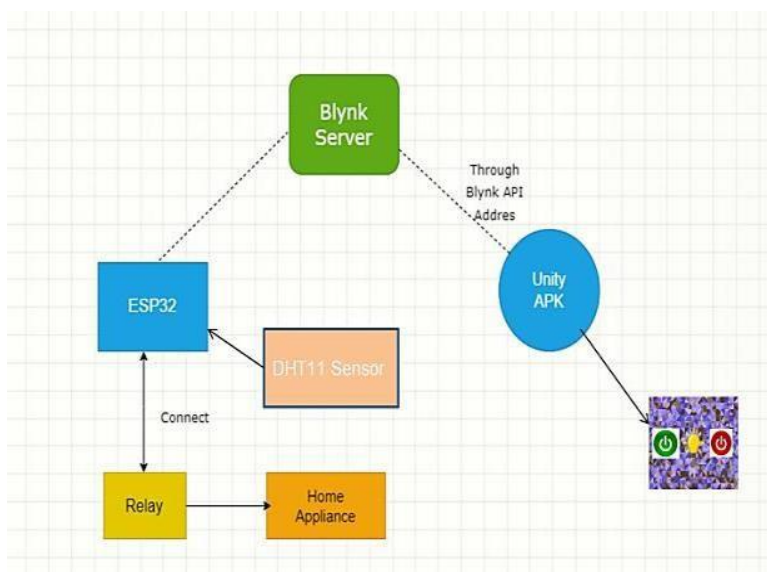


Figure 10. Methodology of Home Automation System.

FUTURE SCOPE

Smarter Home Security Systems give you the opportunity to protect your house and loved ones while also delivering with them the advantages of a contemporary Smart Home. It's more crucial than ever to conserve energy because energy costs are at an all-time high. Additionally to trying to be more environmentally responsible, save money!

Wireless networking systems function in a way that makes it likely that they are already aware of how much energy your property as a whole and specific appliances are using. We believe artificial intelligence will start to have an effect in the future. Smart Energy devices will advance beyond simple energy consumption monitoring to become more foresighted and able to anticipate things like fluctuating energy prices and dynamically modify usage to consume energy when costs are lower.

CONCLUSION

In this work, we suggest an Internet of Things-based security monitoring system. The ESP32, PCB Board, Step Down 12V to 5V, Adaptor 12V, Reset Button, PIR Sensor, Green LED, Red LED, Magnetic Sensor, Internal Touch Sensor, Two MOSFET, Alarm Buzzer, and Electric Strike Door Lock are the components of our suggested system. The Blynk API is tested under various circumstances, and the app is tested over a variety of distances to guarantee the device's mobility. In addition, the NodeMCU is tested over a wider range than the ESP32 and takes longer, so its sensitivity is low. Finally, it was tested with various Wi-Fi networks in various locations with weak Wi-Fi signals, and it required reduced data consumption while still performing optimally. The model created regulates several home appliances over a variety of ranges.

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